MILITARY SPECIFICATION

GENERATOR, SMOKE, MECHANICAL PULSE JET, M3A3

This limited coordination military specification has been prepared by the US Army Armament Research and Development Command, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-G-51066C. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a smoke generator for use in producing smoke screens.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue OF the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

Federal PPP-T-60 - Tape: Packaging, Waterproof.

FSC 1040

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier, Material, Grease Proofed, Waterproofed, Flexible.
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-F-12070 - Fog Oil.

STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-171 - Finishing of Metal and Wood Surfaces.
MS 51938 - Seal, Metallic, Lead and Wire.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 31-15-1259 - Generator, Smoke, Mechanical, Pulse Jet, M3A3, Assembly.
31-15-1259 - Generator, Smoke, Mechanical, Pulse Jet, M3A3.
31-15-1419 - Marking for Generator, Smoke, Mechanical, Pulse Jet, M3A3.
31-15-1420 - Packing for Generator, Smoke, Mechanical, Pulse Jet, M3A3.
31-15-1424 - Packing, Tools and Accessories.

PUBLICATIONS

DEPARTMENT OF THE ARMY

TM-1040-202-12 - Generator, Smoke, Mechanical, Pulse Jet, M3A3.

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(Copies of Specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D3699 - Kerosine.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited in TDPL 31-15-1259 shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the generator shall conform to the specifications and drawings listed on TDPL 31-15-1259.

3.2 Assembly. The generator shall be assembled as shown on Drawing 31-15-1259.

3.3 Functioning.

3.3.1 Smoke generator assembly (Dwg 31-15-1259)0

3.3.1.1 Functioning. The smoke generator shall be operated without stopping or stalling and shall produce smoke continuously for a minimum period of 30 minutes. The fog oil shall be consumed at a minimum rate of 40 gallons per hour (gph) when the oil metering valve is opened *or* adjusted to the proper setting which will produce maximum smoke continuously. Fog oil consumption shall be verified at 15 minute intervals. During production of smoke, excess oil shall flow from the exhaust hose. There shall be no evidence of gasoline or fog oil leakage when tested in accordance with 4.5.4.1(a).

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3.3.1.2 Endurance. The smoke generator shall be operated without stopping or stalling, and shall produce smoke continuously for a minimum period of 4 hours. Minor repairs shall be permitted provided the time does not exceed 15 minutes. Test time for restart shall be from point of stoppage. Total test time shall be counted only when the generator is producing smoke. Down time or repair time shall not be counted. Replacement of parts shall be limited to generator "on board" repair parts. The fog oi1 consumption shall be in accordance with 3.3.1.1 and shall be verified at one hour internals. During production of smoke, excess oil shall flow from the exhaust hose. There shall be no evidence of gasoline or fog oil leakage when tested in accordance with 4.5.4.l(b).

3.4 Preproduction. Prior to the start of production, a preproduction lot of generators shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The generator assembly and components shall be free from damage, contamination and sharp edges that could affect functioning or personnel safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Precautionary.

4.2.1 General precautions. This specification involves materials which are potentially hazardous to personnel. All applicable safety roles, regulations and procedures shall be followed in the handling and processing of these materials.

4.2.2 Functional testing precautions. During the production of fog oil smoke, a protective mask shall be provided and held at "ready" position in the vicinity of dispersed smoke, and is to be worn if exposure appears likely.

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Bathing and laundering of clothing following the production of smoke will reduce the risk of skin irritation following exposure to fog oil smoke. Personnel shall avoid skin contact with fog oil and breathing of fog oil smoke, vapors or fumes.

4.3 Classification of inspections.

(a) Preproductlon inspection (see 4.4)
(b) Quality conformance inspection (see 4.5)

4.4 Preproduction.

4.4.1 Lot. Prior to the start of regular production, two generators shall be produced using the same methods, materials, and equipment that will be used during regular production.

4.4.2 Examination and test. The generators shall be examined and all the requirements of this specification.

4.4.3 Acceptance/relection criteria. The generators shall meet all the requirements of this specification to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.5 Quality conformance inspection.

4.5.1 Lotting. A lot of components or smoke generators shall consist of like components or smoke generators produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.5.2 Sampling.

4.5.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.5.2.2 For endurance. Sampling shall be conducted in accordance with MIL-STD-105, level S-4, using an AQL of 2.5 percent*

4.5.3 Inspection procedure.

4.5.3.1 For examination and nondestructive test. Sample components and smoke generator assemblies, including levels A and B packaging shall be examined in accordance with the classification of defects and with MIL-STD-105. The generator shall be tested for endurance in accordance with 4.5.4.1(b) using an AQL of 2.5 percent defective.

4.5.3.2 For critical defects. Each item in the lot shall be tested for its critical characteristics listed in the classification of defects.

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4.5.3.3 Classification of defects.

(a) Generator, smoke, mechanical, pulse jet, M3A3 (Dwg 31-15-1259).

Categories	Defects	Acceptance standards
Critical:		
1	Functioning	4.5.4.1(a)
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Identification plate missing or incorrect	
103	Finish missing or incorrect	
104	Workmanship (3.5)	
105	Sealing compound missing	
Minor:	AQL 2.5 percent defective	
201	Carrying handle not free	

(b) Packaging inspection (section S).

Categories	Defects
Critical:	None defined
Major	AQL 1.0 percent defective
101	Preservation missing or incorrect
102	Contents missing or incorrect
103	Contents incorrectly located or secured
104	Strap missing, loose, or incorrect
105	Marking incorrect or missing
106	Workmanship (3.5)
107	Damaged areas not repainted
108	Lead seal missing or broken
Minor:	AQL 2.5 percent defective
201	Loose nails

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4.5.4 Tests. Tests shall be conducted as follows:

4,5.4.1 Functioning.

(a) Smoke generator assembly. Operate the smoke generator in accordance with TM 3-1040-202-12 using fog oil, type SGF2, MIL-F-12070. Operate the smoke generator for the specified time period (see 4.2). During the time the generator is continuously producing smoke, determine the fog oil consumption at 15 minute internals. Observe the exhaust hose for flow of excess oil and the complete system for gasoline or fog oil leakage. No leakage is permitted.

(b) Smoke generator endurance. Operate the generator in accordance with TM3-1040-202-12 for the required period of the (see 4.2). During the time the generator is continuously producing smoke, determine the fog oil consumption at l-hour intervals. Observe the exhaust hose for flow of excess oil and the complete system for gasoline or fog oil leakage. No leakage is permitted.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, or B as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Preservative. The following procedures shall be used:

(a) Prime unit. Remove the strainer from the fog oil intake line and replace. Disconnect the air pressure line and flexible fuel line from the engine head assembly. Drain the gasoline from the fuel tank and float bowl assembly through the flexible fuel line. Remove the fuel sediment strainer and replace it. Remove the engine head assembly and replace the engine valve. Re-install the complete engine head assembly. Open all fuel line valves except the float bowl shut off cock and gas tank drain cock. Fill fuel tank with #l fuel oil (ASTM D3699). Open float bowl shut off cock and gas tank drain cock simultaneously and drain system. Close all valves. Reconnect flexible fuel line and air pressure line to engine head assembly. Secure fuel tank cap. Close and latch front cover.

(b) Hinges, draw bolt (latch and catch) and hasp. Treat functional areas with P-9 preservative oil in accordance with MIL-P-116. Work items during application to insure penetration* Wipe off all excess oil.

5.1.1.2 Repainting. Damaged portions of painted surfaces shall be treated and painted in accordance with MIL-STD-171 finish 20.24, color forest green, No. 34079 (Fed. Std. No. 595).

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5.1.1.3 Unit packing.

(a) Strainer guard. Wrap strainer guard assembly in MIL-B-121, type 2, grade A, class 2, greaseproof wrap, size 6 by 12. Wrap so that length of wrap runs with length of hose. Secure wrap to itself in two places and hose in one place using PPP-T-60 tape, size 2 by 6 (3 pcs). Do not attach tape to any metal parts.

(b) Exhaust hose. Plug the exhaust hose using a MIL-C-5501/7 Dash No-F13 plug.

(c) Nozzle(s). Plug the nozzle opening(s) using plugs conforming to MIL-C-5501/7 Dash No-F22.

(d) Accessories, tools, repair parts and mount nuts and bolts. Examine tool box to insure all contents are present, packaged and properly located in accordance with Drawing 31-15-1424. Close latch and secure with a lead seal (MS51938-5). Secure carrying handles, 4 each, in the retracted position with 4 pins by aligning holes.

5.1.2 Level B. Preservation and unit packing shall be as specified in 5.1.1.

5.2 Packing. The level of packing shall be as specified in the contract or purchase order (see 6.1).

5.2.1 Level A. The smoke generator and its accessories, tools, repair parts, and technical publication, preserved and unit packed as specified In 5.1.1 shall be packed as shown on Drawing 31-15-1420.

5.2.2 Level B. The smoke generator and its accessories, tools, repair parts, and technical publication, preserved and unit packed as specified in 5.1.1 shall be packed as shown on Drawing 31-15-1420.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STW129. The shipping container shall be marked in accordance with Drawing 31-15-1419.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification
(b) Level of preservation, packaging and packing required
(c) Preproduction:

(1) Time allowed for contractors to submit samples for Government test and evaluation after award of contract.

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(2) Name and address of test facility and shipping instructions when testing is performed by the Governmet.
(3) Time required for the Government to notify the contractor whether to proceed with production.

6.2 Repair parts packagig. When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts data sheets covering the applicable components of the smoke generator should be used. (Copies of repair parts packaging data sheets may be obtained from Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-E, Aberdeen Proving Ground, MD 21010).

6.3 Assembly. The following specification is used in the assembly of the smoke generator:

MIL-P-51062 - Pump, Fog Oil, Smoke Generator, 144.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous I issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A152

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-G-0051066D(EA) - GENERATOR, SMOKE, MECHANICAL PULSE JET, M3A3

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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